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Abstract #IP38: Does Triplet Therapy Improve Survival in Metastatic Hormone-Sensitive Prostate Cancer? A Bayesian Individual Patient Data Reanalysis

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INTRODUCTION

- Triplet therapy (ADT + docetaxel + ARAT) vs doublet (ADT + ARAT) remains controversial in mHSPC.
- Prior rIPD analyses showed no significant OS benefit, but could not quantify probability of benefit.
- **Objective:**
  To quantify the probability that triplet therapy improves OS using a Bayesian reanalysis of phase III trial data.

METHODS

- **Data:**
  - 8 phase III trials (ARASENS, PEACE-1, ARANOTE, TITAN, ENZAMET, ARCHES, STAMPEDE, LATITUDE)
- **Models:**
  - Bayesian Cox proportional hazards model
  - Hierarchical Poisson GLMM
  - Noninformative prior: Normal(0,10)
  - Sensitivity analysis: weakly skeptical prior Normal(0,0.5)
  - Estimation via MCMC
- **Key endpoints: OS**
  - Pr(HR < 1.0) → any benefit
  - Pr(HR < 0.8) → clinically meaningful benefit
- **Additional:**
  - Tumor volume stratification (high vs low)
  - Treatment class comparison (AR inhibitor–based vs abiraterone-based triplets)
  - Secondary endpoints: rPFS and TTmCRPC

RESULTS

Overall Population

HR: 1.04 (95% CrI 0.92–1.18)

Pr(HR < 1): 36%

Pr(HR < 0.8): 11%

** HR, hazard ratio; CrI, credible interval*

High-volume

HR: 0.89 (95% CrI 0.77–1.03)

Pr(HR < 1): 90%

Pr(HR < 0.8): 19%

** HR, hazard ratio; CrI, credible interval*

Low-volume

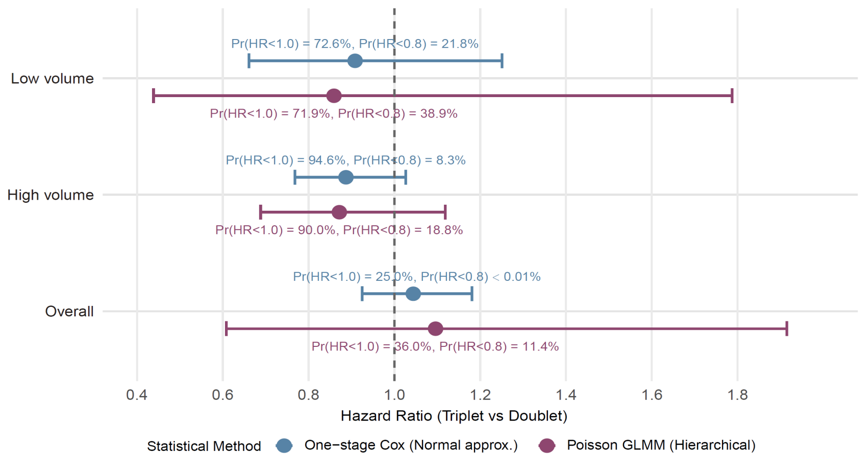
HR: 0.91 (95% CrI 0.66–1.25)

Pr(HR < 1): 73%

Pr(HR < 0.8): 22%

** HR, hazard ratio; CrI, credible interval*

Figure 1. Posterior distributions of overall survival comparing triplet vs doublet therapy. Points represent posterior medians; whiskers indicate 95% CrI. HR < 1 favors triplet therapy.



Triplet therapy shows low probability of meaningful OS benefit overall, with possible modest benefit in high-volume disease.

CONCLUSIONS

ADT + ARAT doublet remains the rational default; triplet therapy may be considered selectively in high-volume patients.

- No clear OS advantage in the overall population
- Modest probability of benefit in high-volume disease
- Probability of clinically meaningful benefit remains low

DISCLOSURES

The authors have nothing to disclose.